6. Optionals and Optional Handling

Task 1:

Code:

```
struct Address {
 let city: String?
}
struct User {
 let name: String
 let age: Int?
 let address: Address?
}
func describe(_ user: User) -> String {
 guard let age = user.age else {
 return "\(user.name): Age is unknown"
 }
 let city = user.address?.city ?? "unknown city"
 return "\(user.name), Age: \(age), City: \(city)"
}
let user1 = User(
 name: "Brinda",
 age: 22,
 address: Address(city: "Perambalur")
)
let user2 = User(
 name: "Deepika",
 age: nil,
 address: Address(city: "Coimbatore")
)
```

```
let user3 = User(
 name: "Ranisha",
 age: 21,
 address: Address(city: nil)
)
print(describe(user1))
print(describe(user2))
print(describe(user3))
let num1 = Int("42")
let num2 = Int("abc")
print(num1 as Any)
print(num2 as Any)
```

OUTPUT:

```
Output

Brinda, Age: 22, City: Perambalur
Deepika: Age is unknown
Ranisha, Age: 21, City: unknown city
Optional(42)
nil
```

Task 2:

Code:

```
let values = ["10", "x", "25", "", "7.5", "40"]
let validNumbers = values.compactMap { Int($0) }
print("Valid integers:", validNumbers)
let sum = validNumbers.reduce(0, +)
print("Sum:", sum)
let invalidCount = values.count - validNumbers.count
```

```
print("Invalid entries:", invalidCount)
let doubled = values.lazy
 .compactMap { Int($0) }
 .prefix(1)
 .map { $0 * 2 }
print("First valid number doubled:", Array(doubled))
```

OUTPUT:

Output

```
Valid integers: [10, 25, 40]
Sum: 75
Invalid entries: 3
First valid number doubled: [20]
```

Task 3:

Correct Code:

```
let scores = ["Asha": 90, "Ravi": 72]
let kiran = scores["Kiran"]
let number = Int("12a")
if let score = kiran {
 if let value = number {
 print(score + value)
 } else {
 print("Invalid number: Cannot convert 12a to an integer")
 }
} else {
 print("Score not found for Kiran")
}
```

OUTPUT:

Output

```
Score not found for Kiran
```